

PRAKRUTHI MANJUNATHA

prakm@umich.edu • 734-489-4536 • [linkedin.com/in/prakruthi-manjunatha](https://www.linkedin.com/in/prakruthi-manjunatha) • Ann Arbor, MI

EDUCATION

University of Michigan

Ann Arbor, MI

Master of Science in Engineering in Biomedical Engineering | GPA: 3.878 / 4.000

May 2025

Coursework: Bioinformatics & Computational Biology, Data Science & Predictive Analysis, Biostatistics.

M S Ramaiah Institute of Technology

Bangalore, India

Bachelor of Engineering in Medical Electronics Engineering | GPA: 3.400 / 4.000

June 2023

Coursework: Digital Signal Processing, Basics of Machine Learning, Biomedical Image Processing, Digital Image Processing.

WORK EXPERIENCE

University of Michigan

Ann Arbor, MI

Research Assistant - Systems and Clinical modeling

Jan 2024 – Present

- Developed a MATLAB pipeline to couple differential gene expression with reaction-level knockouts in genome-scale metabolic models (COBRA), using probabilistic thresholds to simulate flux perturbations.
- Analyzed flux distributions from genome-scale metabolic models and generated heatmaps to visualize pathway activity, enabling comparisons with published literature for biological validation.
- Optimized metabolic models by refining constraints and parameters, which improved pathway consistency and enhanced predictive accuracy in cancer metabolism studies.
- Co-authored a chapter on systems biology modeling in metabolic and epigenetic dysregulation, demonstrating different machine learning methods for biological modeling.

miRcore

Ann Arbor, MI

Group Leader - Computational Biology Summer Camp

June 2025 – Aug 2025

- Mentored groups of high school students through week-long biomedical research projects using transcriptomic datasets.
- Taught computational biology skills including gene expression analysis, biomarker discovery, and introductory machine learning in R.
- Led sessions on core wet lab techniques including PCR, gel electrophoresis, and DNA extraction.
- Conducted interactive discussions and practical training sessions to develop students' skills in scientific reasoning, experimental design, and statistical analysis.

Ross School of Business

Ann Arbor, MI

Data Scientist - Academic Performance Analytics

Aug 2024 - April 2025

- Built predictive models in R (caret, randomForest) and Python (scikit-learn, pandas) to analyze student GPA and demographic data, identifying key predictors of academic performance and informing program improvements.
- Used R (dplyr, ggplot2, tidyverse) to analyze performance bottlenecks in quantitative and communication courses, contributing to curriculum revisions and enhanced student outcomes.
- Developed interactive Sankey diagrams (networkD3, Plotly) to visualize student course transitions and attrition points, supporting advising strategy optimization.
- Conducted statistical analyses, including regression (linear and mixed-effects models) and non-parametric tests (Wilcoxon, Kruskal-Wallis), to assess the effectiveness of admissions rubrics and interventions.

University of Michigan

Ann Arbor, MI

Library Engagement Fellow

Nov 2023 - Aug 2024

- Designed and curated a library exhibit through collaborative research and content development with faculty and staff.
- Created articles and educational materials to support the exhibit and stimulate student interest.
- Organized public programming and events to bring students together and foster active engagement with library resources.

Indian Institute of Science

Bangalore, India

Research Assistant - Computational Biology

Sept 2021 - Jan 2023

- Co-authored a publication in collaboration with Duke Cancer Institute, employing mathematical modeling to investigate the regulatory roles of transcription factors, ELF3 and HOTAIR, in tumor progression and EMT.
- Processed and visualized single-cell RNA-seq cancer data using Bioconductor tools in R and MATLAB, performing t-tests and Spearman correlation to identify significant gene expression differences across clinical subgroups.
- Performed Kaplan-Meier survival analysis using ProgGene and TCGA/CCLE-derived clinical data, stratifying patients by ELF3 and WT1 expression to assess survival outcomes and prognostic potential.

PROJECT EXPERIENCE

University of Michigan

Ann Arbor, MI

Course Project - Project Engineer

Aug 2023 – Dec 2023

- Collected Processed raw ECG signals from the PhysioNet dataset to extract clinically relevant features (RR intervals, QRS duration, HRV metrics) using R.
- Engineered a feature set capturing signal morphology and variability for binary classification (normal vs. abnormal ECG).
- Built and evaluated multiple machine learning models including logistic regression, random forest, and SVM using caret, achieving 86% accuracy.
- Visualized ECG signals and classifier performance to support interpretability and insight into clinical relevance.

M S Ramaiah Institute of Technology

Bangalore, India

Biomedical Engineer

Jan 2023 - May 2023

- Designed and engineered a compact automated variable-resistance wrist rehabilitation device aimed at enhancing upper-limb mobility in stroke patients.
- Applied principles of human biomechanics and device optimization to customize resistance-based therapy protocols.
- Conducted device testing and user-centered design iterations simulating patient use scenarios.
- Awarded Best Project recognition and secured \$6,255 USD in funding for developing a novel at home rehabilitation device.

M S Ramaiah Teaching Hospital

Bangalore, India

Undergraduate Researcher

Sept 2022 - Dec 2022

- Collected synchronized multimodal data, including audio, video, temperature, SpO₂, and heart rate, from infants in the NICU using wearable and contactless sensors.
- Developed a video analysis pipeline to extract facial features for detecting crying events and estimating pain levels in neonates.
- Applied machine learning techniques to classify pain-related events and correlate them with clinical annotations.
- Collaborated with clinicians to align algorithm outputs with clinical pain assessment protocols and real-world NICU constraints.

SKILLS

Languages & Tools: Python, R, MATLAB, SQL, Linux, COBRA, Gurobi, TensorFlow, SciPy, PyTorch, Keras, Pandas & NumPY

PUBLICATIONS

- Prakruthi Manjunatha, Subbalakshmi, A.R., Sahoo, S. *et al.* The ELF3 transcription factor is associated with an epithelial phenotype and represses epithelial-mesenchymal transition. *J Biol Eng* 17, 17 (2023). <https://doi.org/10.1186/s13036-023-00333-z>
- P. Manjunatha., et al., "Automatic Detection of Sleep Stages Using Deep Learning Algorithm," 2022 4th International Conference on Circuits, Control, Communication and Computing (I4C), Bangalore, India, 2022, pp. 436-439, doi: 10.1109/I4C57141.2022.10057685.